SIG-Net: GNN based dropout prediction in MOOCs using Student Interaction Graph

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ABSTRACT

As the COVID-19 pandemic has increased interest in remote learning, so has interest in massive open online courses (MOOCs). MOOCs have experienced rapid growth, offering the advantages of flexible education without time and space constraints. However, a serious challenge in MOOC learning remains high dropout rates; therefore, significant research efforts have been devoted to predicting student dropouts using deep learning as well as machine learning techniques. In this study, we show that graph neural networks (GNNs) can effectively recognize triadic patterns of interactions among students, objects, and courses, producing effective performance outcomes in MOOC dropout prediction. We propose the Student Interaction Graph Network (SIG-Net) model, which extracts subgraphs from a student interaction graph and predicts dropouts by learning student's interactions in the course that the subgraphs contain. In a student interaction graph, subgraphs provide important information about individual students' interactions in a particular course, which is very helpful for dropout prediction. Our model shows that GNNs can effectively learn from student interaction information in the course, and that student interaction information is very powerful in predicting student dropout. We evaluate the proposed model using two real-world datasets: KDDCUP 2015 and NAVER Edwith & Boostcourse. This study is the first dropout prediction study using NAVER Edwith & Boostcourse Our evaluation results show that SIG-Net is effective in learning from student interaction, outperforming the well-known machine learning and deep learning models. The source code is available at https://github.com/Noverse0/SIG-Net.git.

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CCS CONCEPTS

• **Applied computing** → Education; • **Computing methodologies** → Artificial intelligence.

KEYWORDS

MOOC dropout prediction, Graph Neural Network, Education, Remote Learning, Artificial Intelligence

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1 INTRODUCTION

The COVID-19 pandemic has widely affected our way of life, including education. As the pandemic has disturbed traditional educational practices, there has been an increased interest in online education [7, 21], as it can provide high-quality lectures tailored to students' needs without the constraints of time and place. One prominent form of online education is massive open online courses (MOOCs), which are web-based courses open to public and offered via dedicated platforms such as Khan Academy[1], XuetangX[2], Edwith[3] and Boostcourse[4].

While promising as a new form of cost-effective, convenient, and personalized education model, MOOCs have a major problem with high dropout rates; dropout rates in online education tend to be much higher than in traditional in-person education. Studies on MOOCs have shown that most courses have a completion rate below 10% [8]. For example, statistics from a course conducted by the University of California on the Coursera platform indicated that only 7% of students completed the courses. Another study based on the completion data collected for 221 MOOCs showed a completion rate ranging from 0.7% to 52.1%, with a median of 12.6% [12]. The completion rate often depends on the learning method of the

[1]https://www.khanacademy.org/
[2]https://www.xuetangx.com/
[3]https://www.edwith.org/
[4]https://www.boostcourse.org/

MOOC; however, there is generally a much higher rate of students failing to complete the courses compared to traditional in-person education. While this high dropout rate phenomenon presents a limitation of online education, predicting student dropouts in advance creates an opportunity to intervene in time to encourage students to stay in the course [18]. In this study, we define dropout as the absence of any interaction by a student in the course for a specified period. We consider the student's interaction with the course to be the student's log record. We will use interaction log records from the history (past) period to predict dropouts for the prediction (future) period. Figure 1 illustrates the problem definition of whether a student will be drop or non-drop in the prediction period using information about the student's interactions during the history period.

Prior studies on MOOC dropout prediction have encountered several challenges and limitations. First, these studies often relied on complex feature extraction processes, making it difficult to generalize the methods across different datasets. The appropriateness of feature extraction techniques varies depending on the specific dataset, leading to a lack of generic applicability. Additionally, the inclusion of students' personal information in the prediction models can introduce inaccuracies and biases if the information is incomplete or incorrect. A second limitation is the insufficient handling of the intricate relationships between students and course objects. Courses are composed of various objects, and a comprehensive understanding and mastery of each object is crucial for successfully completing a course without dropping out. Prior studies have focused primarily on the types of objects, without considering the nuanced relationships between students and these objects. These studies often aggregate data at the level of object type (e.g., 'video', 'quiz', etc.) rather then considering individual objects. In our approach, we aim to address this limitation by treating individual objects within the course as nodes, allowing us to effectively capture the relationships between students and objects.

To address the aforementioned challenges and limitations, we propose **SIG-Net**, which utilizes graph neural networks (GNNs) for **S**tudent **I**nteraction **G**raphs. GNNs are well suited to learn the relationship information between students and courses because they have a well-defined field for the relationship between users and items [15]. SIG-Net focuses on learning the interactions of students in a course. Our approach avoids the complexity of feature extraction; instead, it focuses on capturing the patterns of student interactions within the course. We construct a heterogeneous graph [25], called the Student Interaction Graph, which consists of three types of nodes: enrollments, objects, and courses. In this graph, an enrollment represents a student enrolled in a course, and an object represents an activity associated with the course. Object nodes are not generated at the level of object type, but rather each object is generated as an individual node. For example, if a course has 10 videos and 5 quizzes, there will be 15 object nodes. This helps to further enrich the interaction information between the student and the course. Generating a node per enrollment is to reflect the common characteristics of multiple students taking a course at a particular time. In other words, there may be certain shared characteristics that students taking the same course exhibit. Generating a node per enrollment is to capture them, which was considered to be more effective for dropout prediction than generating a node per student as a student can exhibit different behaviors depending on courses and time periods.

To handle the continuous enrollment of new students in a course, we have adopted an inductive method [9, 30] that extracts subgraphs based on target enrollments and their corresponding course nodes. Due to the characteristics of MOOC, where enrolled students are constantly changing, it is impractical to learn the entire graph. Three graphs were generated with 10 day intervals to capture the evolving characteristic of student interactions over time. Consequently, a history period of 30 days was utilized and subgraphs were extracted from each of the three graphs. For each extracted subgraph, a relational graph convolutional neural network (RGCN) [22] was used to generate the subgraph embedding, which takes into account the relational nature of the graph to help learn the complex interactions between students and courses.Finally, a Bi-directional long short term memory (Bi-LSTM) [23] was used for the downstream task of predicting student dropouts. In this study, the embeddings of three subgraphs at 10 day intervals are used to predict dropout, but there could be more subgraph embeddings if the train and test periods were adjusted. Therefore, it is important to learn how student interactions change over time. By inputting the subgraph embeddings into the Bi-LSTM in chronological order, we enabled the model to learn the temporal changes in student interactions and to make dropout predictions.

To the best of our knowledge, there have been no existing graph-based studies specifically on the KDDCUP 2015 dataset for dropout prediction. Additionally, our study represents the first dropout prediction study conducted on the NAVER Edwith & Boostcourse dataset. The main contributions of our study are summarized as follows:

- We propose a graph-based method that focuses on interactions of enrolled students in a course for dropout prediction.
- We show that training an evolving graph with subgraphs that change over time is efficient for dropout prediction.
- We show how well and robust our proposed model performs compared to existing dropout prediction models.
- Through experiments with various GNNs methods, we propose a GNNs method suitable for dropout prediction.

2 RELATED WORKS

2.1 Machine Learning-Based Methods

Early research on dropout prediction predominantly employed machine learning (ML)-based methods, as highlighted by Dalipi et al. [3], who found that logistic regression, SVMs, and decision tree models were widely used. For example, Kloft et al. [16] utilized logistic regression and linear SVMs to predict student engagement in the following week. Balakrishnan et al. [1] employed logistic regression in their study, and Chen et al. [2] introduced a novel hybrid algorithm that combined decision trees and extreme learning machines (ELM) without iterative training. Jin [11] proposed an optimized SVR model for predicting student dropout based on the extracted features.

One limitation of ML-based methods is their reliance on complex feature extraction techniques, with the assumption of feature independence, which is not well supported by real-world datasets. Moreover, these methods often fail to capture student interactions

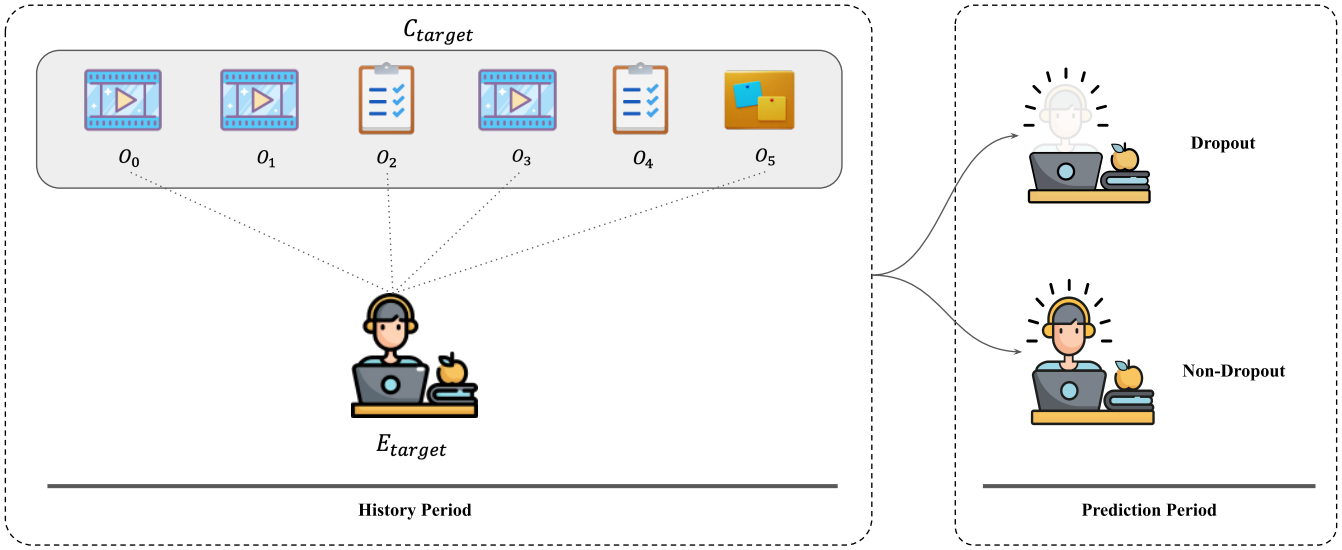

Figure 1: An illustration of MOOC dropout prediction. SIG-Net learns about enrolled students' interactions within a course to make dropout predictions.

within MOOCs and may incorporate non-activity-related information, such as student demographics. Using student demographics information can introduce bias into the model, making it very vulnerable to inaccurate values. Consequently, the performance of ML-based methods heavily relies on the feature extraction approach and may not yield satisfactory results.

2.2 Deep Learning-Based Methods

In recent years, DL-based methods have gained popularity in dropout prediction research. ML-based methods can't capture information about students that changes over time, and their performance is limited because they assume variables are independent. Therefore, many researchers have shifted the focus of MOOC dropout prediction from machine learning methods to deep learning methods. Recurrent neural network (RNN)-based approaches incorporated temporal information to enhance the analysis of students' learning data over time, resulting in a further improvement in dropout prediction performance [4, 28, 29]. Additionally, convolutional neural network (CNN)-based methods enabled the extraction of higher-dimensional feature information from student data, contributing to enhanced overall performance[19, 24, 27]. These early DL methods aimed to extract high-dimensional feature information using CNNs, while utilizing RNNs to effectively capture sequential patterns and dependencies in the input data. More recently, DL-based methods incorporating attention mechanisms have been introduced. In addition to utilizing RNNs and CNNs, these methods further improved performance by taking into account correlations between features and giving higher weights to variables with important information [5, 6, 17, 31]. These approaches demonstrated significant performance improvements compared to ML-based methods.

While DL approaches have gained prominence, there have been relatively few graph-based methods for dropout prediction. Karimi et al. [13] conducted a study using a bipartite graph, attaching behavioral information to edges between student and course nodes to predict course completion and grades in a four-class classification task using the OULAD dataset. Qian Hu et al. [10] proposed an attention-based graph convolutional network model to predict future course grades based on a student's past course attendance, using data from George Mason University.

3 PROPOSED METHOD

In this section, we describe the structure of SIG-Net, which was designed to implement three main steps. Figure 2 illustrates the SIG-Net framework.

First, a subgraph is extracted from the entire graph as 1-hop neighbors based on the target enrollment and target course nodes. A subgraph is extracted from each of the three graphs generated with 10 day interactions, each containing 10 days of interactions in the enrolled course.

Next, GNNs are employed to generate embeddings for the extracted subgraphs. Specifically, we suggest using RGCNs as they can effectively capture the multiple relationships within the graph, enabling a better understanding of student interactions. The embeddings of the target enrollment and target course nodes are concatenated to generate the final subgraph embedding.

Finally, the generated embeddings are sequentially fed into a Bi-LSTM network for dropout prediction. The Bi-LSTM is able to capture temporal patterns and dependencies in the embeddings, allowing for accurate dropout prediction. The output of the Bi-LSTM is passed through a sigmoid function to obtain a probability value, indicating the likelihood of dropout.

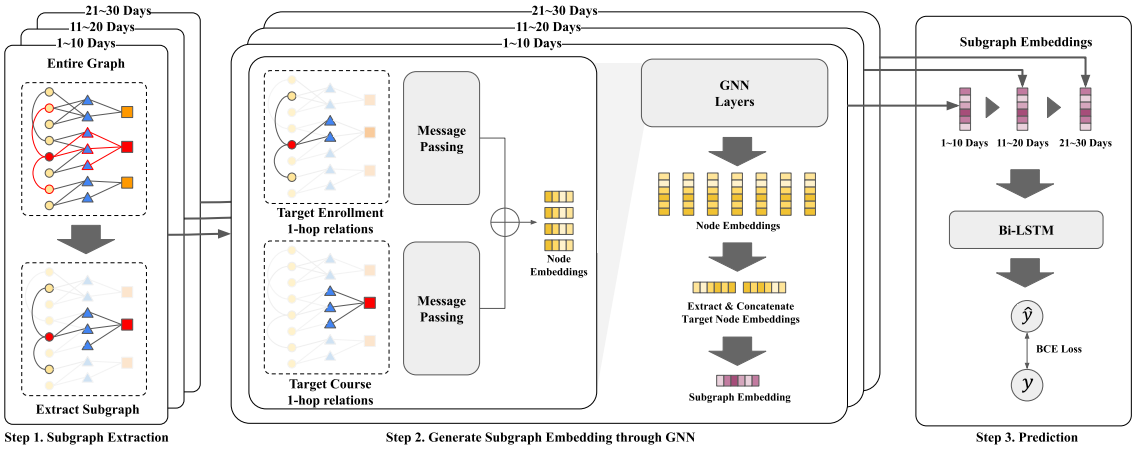

Figure 2: Illustration of SIG-Net Framework. Circles, triangles, and squares represent enrollments, objects, and courses, respectively.

3.1 Graph Construction

We construct a graph to learn the interactions of students enrolled in a course. There are three types of nodes in the graph: Enrollment, Object, and Course. Enrollments represent students enrolled in a course; therefore, the same student enrolled in different courses is treated as a different node. Objects are activities that exist within a course, and every activity is represented by a node. The course node represents the course itself.

There are three types of edges in the graph: Course-Object Edge, Object-Enrollment Edge, and Student Edge. The edge between an object and a course means that the object belongs to that course. Some objects are shared across multiple courses, so they are not necessarily tied to a single course. The edge between an enrollment and an object means that the enrollment is the object with which the interaction existed. An interaction exists if there is a record of access at least once in the log history. The student edge connects enrollments that share the same student. Aggregating interaction information from other courses taken by the same student is helpful for dropout prediction. The student edge contains the dropout status for the next period of enrollment that shares the same student who took different courses. Therefore, the target enrollment node can receive message passing from the student edge with node feature information from the same period for other classes taken by the same student and dropout information for the next period.

The node feature of enrollment is the sum of the daily log records over the history period. The sum of the daily log records is the number of times that a daily log was recorded. Since we are using a history period of 30 days divided into 10 day intervals, the node feature of enrollment will have a vector size of 10. The node features of the course node are the number of objects in the course and the number of students enrolled in the course. The feature is not used for object nodes, and edges associated with object nodes are categorized according to the object type of the object node (e.g., 'video', 'quiz', etc.). These fine-grained edges provide more granular information about the student's interactions, not just the node type, helping the model learn more about the student's interactions.

The total 30 days history period is divided into 10 day intervals and three graphs are generated based on the interactions in that period, each containing the interaction information of the registrations in that period. The 10 day division was chosen because the prediction period is 10 days after the 30 days period. Subgraphs are then extracted from each of the three graphs.

3.2 Subgraph Extraction

To identify relevant interactions and relationships for dropout prediction, a subgraph centered on the target enrollment and the target course is extracted. This subgraph consists of 1-hop neighbors for both the target enrollment and the target course.

For the target enrollment 1-hop neighbors, enrollments that share the same students as the target enrollment and objects that have interacted with the target enrollment are extracted, providing information about interactions with other enrollments and objects within the same course as the target enrollment. For a target course 1-hop neighbors, objects that belong to the target course are extracted, allowing holistic course information to be captured by extracting objects that belong to the target course. By extracting this subgraph, we can focus on specific interactions and relationships related to the target enrollment and the target course.

3.3 Graph Neural Networks

To generate embeddings of the extracted subgraphs using GNNs, specifically RGCNs, we consider the different relationships present in the graph. The relationships are defined in more detail using edges, in addition to the three node types. The Course-Object, Object-Enrollment edges are separated by the type of object. The Student Edge is separated by whether the target enrollment node's neighbor enrollment node indicates a dropout in the next period.

RGCN is an effective GNN method for learning these diverse relationships in the graph. The RGCN layer aggregates the embeddings of neighboring nodes based on the respective edge types and updates the embeddings of the target nodes, enabling us to learn

the complex triadic interactions between students, courses, and objects in the graph.

Among the embeddings generated by RGCN, we concatenate the embeddings of the target enrollment and target course nodes. This combined embedding captures the interaction patterns between the student and the course objects over a specific history period. This subgraph embedding serves as a comprehensive representation of the student's interactions within the course. Equation (2) shows how the RGCN layer aggregates the neighbor node embeddings and updates the target node embeddings as,

$$h_i^{(l+1)} = ReLU\left(W_0^{(l)} h_i^{(l)} + \sum_{r \in R} \sum_{j \in \mathcal{N}_i^r} \frac{1}{|\mathcal{N}_i^r|} W_r^{(l)} h_j^{(l)}\right) \quad (1)$$

where $h_i^{(l)}$ denotes the embedding of node i from the $l-th$ layer and $\mathcal{N}_i^r$ denotes the set of neighbors of node i under relation r. The Rectified Linear Unit ($ReLU$) is adopted as the activation function.

$$S_t^* = concat[h_{e_{target}}, h_{c_{target}}] \quad (2)$$

Among the embeddings obtained from GNN, we concatenate the embeddings of the target enrollment and target course nodes to generate a subgraph embedding S_t^* for the history period t. $h_{e_{target}}$ denotes the embedding of the target enrollment node and $h_{c_{target}}$ denotes the embedding of the target course. These concatenated embeddings capture the combined information of the student and the course they are enrolled in, providing a comprehensive representation of the subgraph.

3.4 Prediction

To predict dropouts, a Bi-LSTM model is used, which takes the embeddings extracted from the three subgraphs as input. The Bi-LSTM model is specifically designed to capture temporal information and sequential patterns in the data.

By feeding the subgraph embeddings into the Bi-LSTM model, it can learn how the interactions of students enrolled in the course change over time. The Bi-LSTM model processes the embeddings in both forward and backward directions, allowing it to capture dependencies and patterns in the temporal sequence of subgraphs, enabling the model to understand the temporal dynamics of student interactions and make predictions about dropouts based on the learned temporal patterns. Equation (3) show how the LSTM uses the past timestep $t-1$ to update the subgraph embedding information to the latest timestep t in each direction:

$$h = LSTM(S_t^*, h^{(l-1)}) \quad (3)$$

S_t^* is the subgraph embedding at timestep t. $h^{(l-1)}$ is the hidden state from the previous step.

The output of the Bi-LSTM at time t is the concatenated values of the forward and backward neural networks, which is expressed in Equation (4). h_t is mapped to probability values between 0 and 1 using a sigmoid function:

$$h = [h_{forward}, h_{backward}] \quad (4)$$

$$\hat{y} = \sigma(h) \quad (5)$$

σ denotes a sigmoid function. $\hat{y}$ closer to 1 indicates a higher likelihood of dropout, while a value closer to 0 indicates a lower likelihood.

4 EXPERIMENTS

In this section, we describe the experiments conducted to evaluate our proposed model in comparison with other extant models while answering the following research questions.

- **RQ 1.** Does SIG-Net perform better than other models in predicting dropouts?
- **RQ 2.** How does each of the graph components affect performance?
- **RQ 3.** How does SIG-Net perform in different history period days and patterns?

4.1 Baseline

Using the same set of features, we compared SIG-Net with four ML models: Random Forest(RF), eXtreme Gradient Boosting(XGB), Support Vector Machines(SVM), and Logistic Regression(LR). Furthermore, our comparison included four Deep learning models: Bi-LSTM, CNN, CNN+LSTM [24], and CNN+Attention+LSTM [31]. The Bi-LSTM model combined node features from both enrollment and course nodes for the graphs of 1-10 days, 11-20 days, and 21-30 days as its input. On the other hand, the CNN model used the same features as the Bi-LSTM but structured them in a 3-row matrix to account for the different time intervals, where each row corresponds to the features used in the graphs of 1-10 days, 11-20 days, and 21-30 days, respectively. CNN+LSTM utilized features extracted by CNN, which were then used as inputs for the LSTM model to make predictions. CNN+Attention+LSTM employed CNN to generate local feature representations and further learned correlations with other features through self-attention mechanisms.

4.2 Datasets

We conducted experiments using two real-world datasets: KDDCUP 2015 (KDDCUP) and NAVER EdWith & BoostCourse (NAVER).

The KDDCUP dataset consists of a total of 39 courses, all in instructor-paced mode (IPM), with a fixed duration of 16 weeks. In IPM classes, students follow a predetermined course schedule, and progress through the course according to the instructor's timeline. This dataset represents a more structured and time-bound learning environment.

In contrast, the NAVER dataset comprises 35 courses in a self-paced mode (SPM). In SPM, students have the flexibility to enroll in the course and start their studies at any time. They can progress through the course materials at their own pace and adjust their learning schedule according to their convenience. This dataset reflects a more flexible and adaptable learning environment.

It is important to note that in SPM, where students have more autonomy in managing their learning journey, the dropout rate is generally higher as observed in the NAVER dataset. The varying dropout rates between the two datasets provide an interesting contrast for evaluating the performance of our dropout prediction model. Figure 3 provides an overview of the dropout ratios for each dataset.

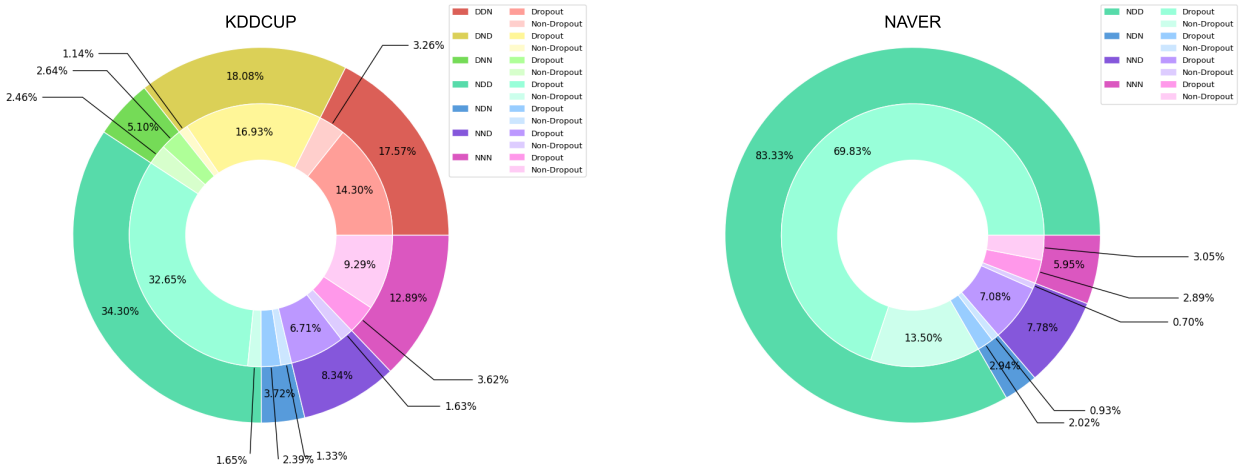

Figure 3: The ratio of enrollments for each pattern when the history period is divided into 10 days and categorized into dropouts and non-dropouts. D means Dropout, N means Non-Dropout.

Table 1: Dropout ratio of datasets

Dataset	Class	Dropout	Non-Dropout
KDDCUP	Total	159,223 (79.25%)	41,681 (20.75%)
	Train	95,581 (79.29%)	24,961 (20.71%)
	Test	63,642 (79.19%)	16,720 (20.81%)
NAVER	Total	127,732 (93.16%)	9,381(6.84%)
	Train	76,645(93.17%)	5,622 (6.83%)
	Test	51,087 (93.15%)	3,759 (6.85%)

The KDDCUP dataset came with pre-separated train and test sets in a ratio of 6:4. Mirroring the same ratio, we split the NAVER dataset into the train and test sets

Table 2: Performance of SIG-Net and the baseline models

	KDDCUP		NAVER	
	AUC	F1	AUC	F1
SIG-Net	**88.87±0.04**	**92.52±0.04**	**88.16±0.08**	**97.28±0.02**
RF	75.05±0.08	91.41±0.02	70.21±0.11	97.13±0.02
XGB	74.81±0.00	92.10±0.00	69.70±0.00	97.18±0.00
SVM	72.02±0.00	91.75±0.00	63.55±0.00	97.07±0.00
LR	69.54±0.00	91.35±0.00	59.95±0.00	96.82±0.00
Bi-LSTM	86.63±0.06	92.11±0.04	87.35±0.03	97.25±0.01
CNN	86.73±0.07	92.09±0.05	86.85±0.12	97.21±0.02
CNN+LSTM	86.68±0.06	92.01±0.03	86.83±0.13	97.17±0.03
CNN+Attention+LSTM	86.56±0.06	92.07±0.06	86.95±0.06	97.21±0.03

We used the same features used for node features when experimenting with the baselines.

4.3 Evaluation Metric

Table 1 shows that, in both datasets, the ratio of dropouts to non-dropouts is not balanced, especially in the NAVER dataset (almost 93% dropout ratio). In this case, it can be misleading if we use accuracy as a metric to evaluate the model. Therefore, we use Area Under the ROC (AUC) and an F1-score for evaluation. We repeated the experiment a total of five times and recorded the corresponding mean values and standard deviations.

4.4 Experiment settings

We implemented the model using PyTorch [5] and DGL [6]. We used Binary Cross Entropy loss and Adam optimizer with an initial learning rate of 0.001. The batch size was fixed at 256, and the embedding size in the hidden layer of the GNN was fixed at 64. We used 4 epochs for the KDDCUP dataset and 7 epochs for the NAVER dataset. Per epoch, we train as many times as the number of enrollment nodes in the train set, which requires more epochs of training because the NAVER dataset has a smaller number of enrollment nodes in the train dataset than the KDDCUP dataset.

[5]https://pytorch.org/
[6]https://www.dgl.ai/

4.5 RQ1

Table 2 presents the performance outcomes of SIG-Net and the baseline models obtained against the KDDCUP and NAVER datasets. The baselines utilized the node features used in the proposed SIG-Net method. The results show that SIG-Net consistently outperforms the baselines on both datasets, without exception.

On the KDDCUP dataset, SIG-Net achieves a 2.14% higher AUC compared to the best performing baseline model, CNN. Similarly, on the NAVER dataset, SIG-Net shows a 0.81% higher AUC compared to the best performing baseline model, Bi-LSTM. These improvements in AUC indicate that SIG-Net effectively captures the student's interaction information by segmenting it differently from the existing baselines. There were clearly noticeable performance differences observed between SIG-Net and the ML-based methods used in the baselines on both datasets, highlighting the advantage of SIG-Net in learning the intricate student interaction patterns and its ability to accurately predict student dropouts.

4.6 RQ2

Table 3 presents performance of alternative SIG-Net configurations, based on different graph structures. In particular, both datasets still have the highest performance compared to the baselines in Table

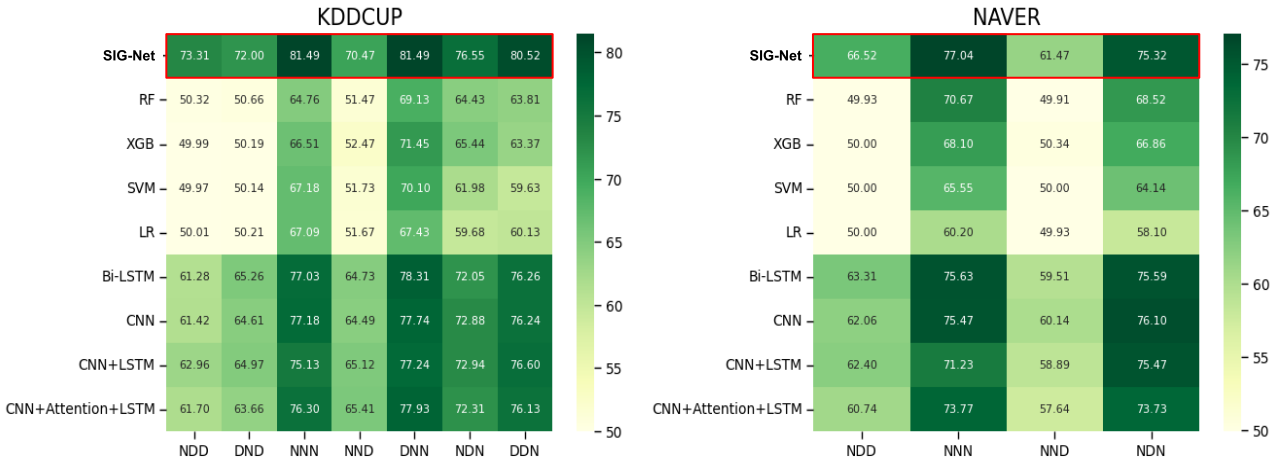

Figure 4: Heatmap of AUC by Histroy period pattern. Darker green colors indicate higher performance.

Table 3: Performance of alternative SIG-Net configurations

	KDDCUP		NAVER	
	AUC	F1	AUC	F1
SIG-Net	**88.87±0.04**	**92.52±0.04**	**88.16±0.08**	97.28±0.02
w/o node feature	88.18±0.04	92.25±0.05	87.64±0.18	97.28±0.02
w/o course node	88.57±0.03	92.43±0.04	87.89±0.11	97.26±0.02
w/o student edge	87.78±0.03	92.40±0.09	87.78±0.09	**97.29±0.02**

The meaning of w/o is without. In the w/o course node case, we utilized a bipartite graph with enrollment node and object node, and used the embedding of the target enrollment node.

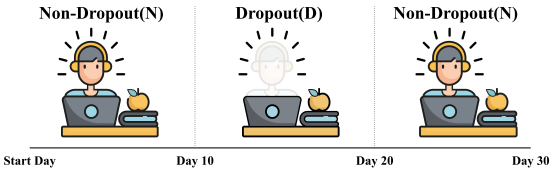

Figure 5: Example of NDN case.

2 even when without node features, indicating that learning the structure with subgraphs is very effective in predicting dropouts. It means that the subgraph structure captures the student interaction information well, and GNN subsequently learns it effectively. In addition, connecting objects belonging to courses via course nodes and representing enrollments sharing the same student using student edges also showed significant performance improvements. In particular, it was meaningful to see the same period interaction information in other courses taken by the same student via student edges. Even though students take different courses, there is a unique pattern for each individual student, emphasizing the importance of capturing both the patterns of different students within each course and the unique patterns of each student.

There were also performance differences in the ablation study between the KDDCUP and NAVER datasets, which we believe is due to the differences between the datasets, as shown in Table 1 and Figure 3. Because the KDCCUP dataset is for IPM courses and the NAVER dataset is for SPM courses, most of the cases in the NAVER dataset are dropout cases even if student edges were used.

4.7 RQ3

We analyzed patterns over history periods to compare the results. Enrolled students can dropout and return to class during a history period. Therefore, we divided the history period into 10 days segments and categorized each segment as either dropout or non-dropout, and examined the AUC according to the patterns created by the three periodic segment combinations. For example, as shown in Figure 5, some students may have log records from 1-10 days, then no log records from 11-20 days, and then again log records from 21-30 days. We define this case as Non-dropout(N), Dropout(D), and Non-dropout(N) case, respectively. We analyzed the results of KDDCUP by classifying them into Non-Dropout (N) and Dropout (D). As for the NAVER dataset, because the enrollment starts when a student enrolls in a course, the first history period always represents a non-dropout case. Therefore, we fixed the first period as non-dropout and categorized the remaining periods into four patterns: NDD, NNN, NND, and NDN. The distribution of each pattern in the datasets can be observed in Figure 3.

According to the results shown in Figure 4, SIG-Net consistently outperforms the baselines across all patterns. In particular in the NDD, NND, and DND cases where predicting the next dropout is challenging, the ML-based baselines struggle to make accurate predictions. However, even in these difficult cases, SIG-Net demonstrates superior performance compared to the baselines. The difficulty in predicting NDD, NND, and DND cases is due to the imbalance and lack of datasets. Figure 3 shows that NDD and NND are very unbalanced in non-dropout and dropout results. DND cases were very rare compared to other cases. In the DND cases, the baselines performed poorly, but because SIG-Net learns students' interactions via graph modeling, it was able to learn even

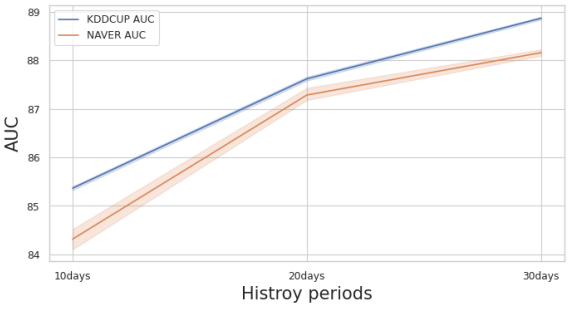

Figure 6: The AUC over the history period. The history period from before the prediction period.

with an unbalanced or insufficient dataset and achieved higher performance.

We further conducted experiments to investigate the effect of limiting the history period on dropout prediction. Figure 6 illustrates the AUC values obtained by dividing the history period into 10, 20, and 30 days prior to the prediction period. In the case where only the last 10 days of the prediction period were used, we were unable to utilize Bi-LSTM and instead employed a fully connected layer for dropout prediction. The best results were obtained when using the entire 30 day history period, indicating that it is important not only to learn the interaction of enrollment, but also to have a good understanding of how the interaction changes in each period. We also found that the difference in performance between using only the last 10 days and 20 days was greater than using 20 days and 30 days, suggesting that interaction information from periods closer to the prediction period is more important for prediction.

5 CONCLUSIONS

In this study, we employed a GNN-based approach to predict dropouts in MOOCs by extracting subgraphs from the enrollment interaction graph. Our proposed model outperformed other baseline methods and demonstrated robustness across various scenarios, highlighting the effectiveness of GNNs in capturing student-course interactions, similar to their success in modeling user-item relationships. Our approach has the potential to be applicable to other tasks beyond dropout prediction that involve predicting student behaviors based on their interactions.

As future work, it would be valuable to explore an explainable model that predicts dropouts based on student interactions; if we could know which of the many factors contribute to each student's dropout, we could use the information to prevent dropouts [14] at the individual level. Furthermore, given the high dropout rate observed in the SPM dataset employed in our study, it would be beneficial to further enhance the learning of non-dropout cases through data augmentation techniques.

As MOOCs are likely to continue growing even beyond the COVID-19 pandemic, reducing the dropout rate among students will remain an important challenge. While our study focused on predicting dropouts, future research could explore strategies to prevent dropouts by recommending appropriate next learning steps for students predicted to be at risk, thereby facilitating their course completion [20, 26]. The current study is an essential step toward that direction.

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